General Science

A Comprehensive Introduction

Topics Covered:

- The Scientific Method
- Matter and Its Properties
- Forces and Motion
- Energy
- Waves and Sound
- Light and Optics
- Electricity and Magnetism
- The Solar System
- Living Things

Chapter 1: The Scientific Method

Science is a systematic way of learning about the natural world through observation and experimentation.

Steps of the Scientific Method

1. **Observation** – Noticing something in the natural world that sparks curiosity.
2. **Question** – Forming a clear, testable question based on the observation.
3. **Hypothesis** – Making an educated prediction that can be tested.
4. **Experiment** – Designing and conducting a controlled test.
5. **Data Collection** – Recording measurements and results accurately.
6. **Analysis** – Interpreting the data to find patterns or trends.
7. **Conclusion** – Determining whether the hypothesis was supported.

> **Key Term:** A *hypothesis* must be falsifiable — it must be possible to prove it wrong through testing.

Variables in Experiments

- **Independent variable** – What the experimenter changes on purpose.
- **Dependent variable** – What is measured as a result.
- **Controlled variables** – Everything else kept constant.

Chapter 2: Matter and Its Properties

Matter is anything that has mass and takes up space. Everything around us is made of matter.

States of Matter

- **Solid** – Fixed shape and volume; particles closely packed and vibrate in place.
- **Liquid** – Fixed volume but no fixed shape; particles slide past each other.
- **Gas** – No fixed shape or volume; particles move freely and rapidly.
- **Plasma** – Superheated ionised gas; found in stars and lightning.

Physical and Chemical Properties

- **Physical properties** can be observed without changing the substance (colour, density, melting point).
- **Chemical properties** describe how a substance reacts to form new substances (flammability, reactivity).

Elements, Compounds, and Mixtures

An **element** is a pure substance made of one type of atom. A **compound** forms when two or more elements bond chemically (e.g., water H_2O). A **mixture** combines substances that are not chemically joined.

Chapter 3: Forces and Motion

A force is a push or pull on an object. Forces cause objects to start moving, stop, or change direction.

Newton's Three Laws of Motion

1st Law (Inertia): An object at rest stays at rest unless acted upon by an unbalanced force.

2nd Law: Force = mass x acceleration ($F = ma$).

3rd Law: For every action there is an equal and opposite reaction.

Types of Forces

- **Gravity** – Pulls objects toward each other; gives weight to matter.
- **Friction** – Opposes motion between surfaces in contact.
- **Normal force** – Perpendicular support from a surface.
- **Tension** – Force transmitted through a rope or cable.

Speed, Velocity, and Acceleration

Speed is distance per unit time. Velocity is speed with direction. Acceleration is the rate of change of velocity.

Chapter 4: Energy

Energy is the ability to do work. It cannot be created or destroyed — only converted (Law of Conservation of Energy).

Forms of Energy

- **Kinetic Energy** – Energy of motion.
- **Potential Energy** – Stored energy due to position or condition.
- **Thermal Energy** – Total kinetic energy of particles in a substance.
- **Chemical Energy** – Stored in chemical bonds (food, fuel, batteries).
- **Electrical Energy** – Energy of moving electric charges.
- **Nuclear Energy** – Released from atomic nuclei in fission or fusion.
- **Radiant Energy** – Carried by electromagnetic waves (light, X-rays).

Energy Transformations

Energy continuously transforms between types. A car engine converts chemical energy into thermal and then mechanical (kinetic) energy. A solar panel converts radiant energy into electrical energy.

> **Did You Know?** Only ~25% of petrol energy becomes motion. The remaining 75% is lost as heat.

Chapter 5: Waves and Sound

A wave transfers energy from one place to another without transferring matter.

Types of Waves

- **Transverse waves** – Particles move perpendicular to the wave's direction (e.g., light).
- **Longitudinal waves** – Particles move parallel to the wave's direction (e.g., sound).

Wave Properties

- **Wavelength** – Distance between two consecutive crests.
- **Frequency** – Number of wave cycles per second (Hertz).
- **Amplitude** – Maximum displacement from rest; relates to energy.
- **Wave speed** – v = frequency x wavelength.

Sound

Sound is a longitudinal mechanical wave requiring a medium to travel. It cannot travel through a vacuum. Speed in air is ~343 m/s. Pitch is determined by frequency; loudness by amplitude.

Chapter 6: Light and Optics

Light is a transverse electromagnetic wave that travels through a vacuum at ~3×10^8 m/s.

The Electromagnetic Spectrum

Radio waves — Microwaves — Infrared — Visible light — Ultraviolet — X-rays — Gamma rays (increasing frequency).

Reflection and Refraction

- **Reflection** – Light bounces off a surface. Angle of incidence = angle of reflection.
- **Refraction** – Light bends passing between media due to a change in speed.
- **Dispersion** – White light splits into a colour spectrum through a prism.

Lenses and Mirrors

- **Concave mirror** – Converges light; used in telescopes and headlights.
- **Convex mirror** – Diverges light; wider field of view (rear-view mirrors).
- **Convex lens** – Converges light; used in magnifying glasses and cameras.
- **Concave lens** – Diverges light; corrects short-sightedness.

Chapter 7: Electricity and Magnetism

Electricity is the flow of electric charge. Magnetism is produced by moving charges. Together they form electromagnetism.

Electric Circuits

- **Current (I)** – Flow of charge, measured in Amperes.
- **Voltage (V)** – Potential difference driving current, measured in Volts.
- **Resistance (R)** – Opposition to current, measured in Ohms.

Ohm's Law: V = I x R | I = V / R | R = V / I

Series vs Parallel Circuits

In a **series circuit**, components connect end-to-end; if one fails all fail. In a **parallel circuit**, components share the same voltage; if one fails others continue. Home wiring uses parallel circuits.

Magnetism

Every magnet has a north and south pole. Like poles repel; unlike poles attract. Electromagnets use electric current through a coil and are used in motors, generators, and MRI machines.

Chapter 8: The Solar System

Our solar system consists of the Sun, eight planets, dwarf planets, moons, asteroids, and comets, all bound by gravity.

The Eight Planets (in order from the Sun)

Planet	Type	Key Feature
Mercury	Terrestrial	Closest to Sun; extreme temperature swings
Venus	Terrestrial	Hottest planet; dense CO2 atmosphere
Earth	Terrestrial	Only known planet with life
Mars	Terrestrial	Red planet; largest volcano in the solar system
Jupiter	Gas Giant	Largest planet; famous Great Red Spot storm
Saturn	Gas Giant	Stunning ring system of ice and rock
Uranus	Ice Giant	Rotates on its side; blue-green colour
Neptune	Ice Giant	Strongest winds in the solar system

The Sun

The Sun is a medium-sized star composed mainly of hydrogen and helium. Nuclear fusion in its core releases enormous energy as light and heat. It accounts for 99.86% of the solar system's total mass.

Chapter 9: Living Things

Biology is the study of life. All living organisms share fundamental characteristics that distinguish them from non-living matter.

Characteristics of Living Things

- **Organisation** – Made of one or more cells.
- **Metabolism** – Carry out chemical reactions to obtain and use energy.
- **Growth** – Increase in size or number of cells over time.
- **Response** – React to changes in their environment.
- **Reproduction** – Produce offspring and pass on genetic information.
- **Homeostasis** – Maintain a stable internal environment.

Cells: The Building Blocks of Life

The cell is the smallest unit of life. **Prokaryotic cells** (bacteria) have no nucleus. **Eukaryotic cells** (plants, animals, fungi) have a membrane-bound nucleus. Key organelles: nucleus (control), mitochondria (energy), ribosomes (protein synthesis).

Classification of Life

Life is classified into three domains: Bacteria, Archaea, and Eukarya. Within Eukarya there are five kingdoms: Animals, Plants, Fungi, Protists, and Monera. Classification helps scientists study relationships and understand evolution.

Summary: Science connects every aspect of our world — from subatomic particles to the vast cosmos. Understanding these principles builds the foundation for all advanced scientific study.